

		not enough energy to get close to the nucleus	1	[2]
3	(a)	(i) reasonable shape as 'inverse' of k.e. line	1	
		(ii) straight line, parallel to x-axis at 15 mJ	1	[2]
	(b)	<i>either</i> (max) kinetic energy ($= \frac{1}{2} mv^2$) $= \frac{1}{2} m \omega^2 a_0^2$	1	
		$15 \times 10^{-3} = \frac{1}{2} \times 0.15 \times \omega^2 \times (5.0 \times 10^{-2})^2$	1	
		$\omega = 8.9(4) \text{ rad s}^{-1}$	1	
		or (k.e. $= \frac{1}{2} mv^2$), $v = 0.44(7) \text{ m s}^{-1}$	1	
		$\omega = v/a = (0.447)/(5.0 \times 10^{-2})$	1	
		$\omega = 8.9(4) \text{ rad s}^{-1}$	1	[3]
	(c)	(i) <i>either</i> loss of energy (from the system) <i>or</i> amplitude decreases	1	
		<i>or</i> additional force acting (on the mass)	1	
		<i>either</i> continuous/gradual loss <i>or</i> force always opposing motion	1	[2]
		(ii) <i>either</i> (now has 80% of its) p.e./k.e. = 12 mJ <i>or</i> loss in k.e. = 3 mJ	1	
		new amplitude = 4.5 cm (allow $\pm 0.1 \text{ cm}$)	1	[2]

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